

CPTR 242 - Semester Project: Algorithms Through Worlds

1. Team Members

Solo Project - Caleb Osorio

2. Chosen Theme

Theme 3 - Spatial Partitioning - asteroid/star field with collision

Dataset:

<https://www.kaggle.com/datasets/ary0nk/spotify-songs-dataset>

Plot every Spotify track as a star in 2D space where X = danceability and Y = energy. Use a custom hash table with spatial hashing for $O(1)$ average-time neighbor ("collision") queries instead of brute-force $O(n)$. Click any star → see its sonic neighbors + song info.

3. Core data structures

- HashTable (custom chaining hash map – no Python dict allowed for storage)
- SpatialHash (uses the custom HashTable to bucket tracks into grid cells)
- Track dataclass (id, name, artists, danceability, energy, + optional third feature for color)

4. Language and tech stack.

Python 3.11+
pandas (data loading)
pygame (rendering + interaction)
numpy (optional fast rendering)

Language: Python

5. Github repo link

<https://github.com/Seahawks1013/spotify-nebula>

6. Rough weekly plan

Week 1 (Apr 20): Proposal approved, GitHub repo set up, dataset downloaded, basic hash table skeleton working with 10K tracks, simple Pygame window showing dots.

Week 2 (May 1): Full CRUD operations implemented in the hash table, naive approach benchmarked.

Week 3 (May 8): Load all ~1.2M tracks into the hash table, basic starfield rendering with spatial grid, 1M element test passing.

Week 4 (May 15): Spatial hashing + grid visualization (draw bucket cells), mouse click for Read and simple “collision” neighbor search.

Week 5 (May 22): Full benchmarks at 10K / 100K / 1M elements + complexity analysis write-up.

Week 6 (May 29): Polish camera movement, song info popups, final reflection and demo preparation.

NOTE: This weekly plan was created/modified using Grok.

7. Stretch goals

- Color stars by a third feature
- Adding song previews (30-second previews)
- Adding a link to the song on Spotify when they hover over it on the Nebula.
- Creating a bio for artists that includes the artists' age, Pictures, and links to their social media.